

# FIITJEE

## ALL INDIA TEST SERIES

### FULL TEST – III

**JEE (Main)-2022**

**TEST DATE: 04-01-2022**

**Time Allotted: 3 Hours**

**Maximum Marks: 300**

**General Instructions:**

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

**Section-A (01 – 20, 31 – 50, 61 – 80)** contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

**Section-B (21 – 30, 51 – 60, 81 – 90)** contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

# Physics

## PART – A

### SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

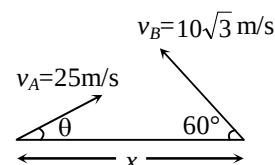
1. What are the dimensions of electrical permittivity?

- (A)  $ML^{-2} T^{-2} Q^{-2}$
- (B)  $M^{-1} L^2 T^{-3} Q^{-1}$
- (C)  $M^{-1} L^{-3} T^2 Q^2$
- (D)  $M^{-1} L^3 T^{-2} Q^{-2}$

Ans. C

Sol. Permittivity  $\epsilon = \frac{(\text{charge})^2}{\text{force} \times (\text{distance})^2} = \frac{Q^2}{MLT^{-2} \times L^2} = M^{-1} L^{-3} T^2 Q^2$

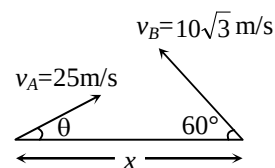
2. Two particles A and B are projected simultaneously in the directions shown in figure with velocities  $v_A = 25\text{m/s}$  and  $v_B = 10\sqrt{3}\text{m/s}$ . If they collide in air after 2s, the angle  $\theta$  is



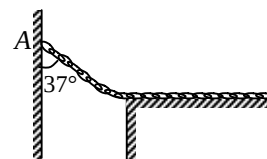
- (A)  $30^\circ$
- (B)  $45^\circ$
- (C)  $53^\circ$
- (D)  $37^\circ$

Ans. D

Sol. For collision,  
 $v_A \sin \theta = v_B \sin 60^\circ$   
 $25 \sin \theta = 10\sqrt{3} \times \frac{\sqrt{3}}{2}$   
 $\sin \theta = \frac{3}{5}$   
 or  $\theta = 37^\circ$



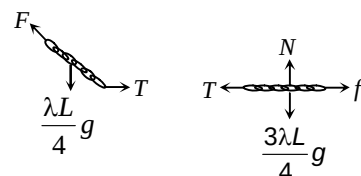
3. A uniform chain of length  $L$  has one of its end attached to the wall at point A, while  $\frac{3L}{4}$  of the length of the chain is lying on table as shown in figure. The minimum co-efficient of friction between table and chain so that chain remains in equilibrium is



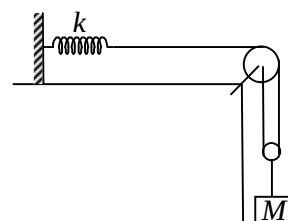
- (A)  $1/3$   
 (B)  $1/4$   
 (C)  $3/4$   
 (D)  $1/5$

Ans. B

Sol.  $F \cos 37^\circ = \frac{\lambda L}{4} g$   
 (where  $\lambda$  is the mass/length of the chain).  
 $F \sin 37^\circ = T = f \leq \mu N$   
 $\Rightarrow \mu \geq \frac{1}{4} \Rightarrow \mu_{\min} = \frac{1}{4}$



4. A block of mass  $m$  is released from rest when the extension in the spring is  $x_0$ . The maximum downward displacement of the block is



- (A)  $\frac{Mg}{2k} - x_0$   
 (B)  $\frac{Mg}{2k} + x_0$   
 (C)  $\frac{2Mg}{k} - x_0$   
 (D)  $\frac{2Mg}{k} + x_0$

Ans. A

Sol. If block M moves a distance of  $x$  the extension in spring increases by  $2x$ .  
 By work energy theorem,

$$\frac{1}{2} k [x_0^2 - (x_0 + 2x)^2] + Mgx = 0$$

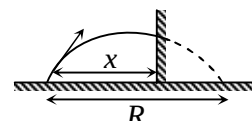
$$-\frac{1}{2}k(4x^2 + 4xx_0) + Mgx = 0$$

$$\frac{1}{2}k(4x + 4x_0) = Mg$$

$$x + x_0 = \frac{Mg}{2k}$$

$$x = \frac{Mg}{2k} - x_0$$

5. A particle is projected with such a velocity that its range is  $R$ . The particle undergoes elastic collision with a wall placed at a distance of  $x$  ( $> R/2$ ). The horizontal distance of point of projection and point where particle hits the ground is

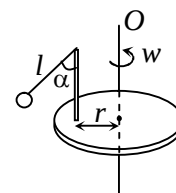


- (A)  $R + x$   
 (B)  $x - R$   
 (C)  $2x - R$   
 (D)  $R + 2x$

Ans. C

Sol. Since collision is elastic, the horizontal velocity reverses its direction.

6. A ball of mass  $m$  is attached to the end of a thread fastened to the top of a vertical rod which is fitted to a horizontally revolving round table as shown. If the thread forms an angle  $\alpha$  with the vertical, the angular velocity  $\omega$  of table is



- (A)  $\sqrt{\frac{g}{l \cos \alpha}}$   
 (B)  $\sqrt{\frac{r}{g \tan \alpha}}$   
 (C)  $\sqrt{\frac{g \tan \alpha}{r + l \sin \alpha}}$   
 (D)  $\sqrt{\frac{g \tan \alpha}{r}}$

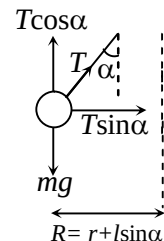
Ans. C

Sol.  $T \sin \alpha = m\omega^2 R$

$$T \cos \alpha = mg$$

$$\tan \alpha = \frac{\omega^2 R}{g}$$

$$\omega = \sqrt{\frac{g \tan \alpha}{R}} = \sqrt{\frac{g \tan \alpha}{r + l \sin \alpha}}$$



7. An artificial satellite of mass  $m$  is moving in a circular orbit at a height equal to the radius  $R$  of the earth. Suddenly due to internal explosion the satellite breaks into two parts of equal masses. One part of the satellite stops just after the explosion. The increase in the mechanical energy of the system (satellite + earth) due to explosion will be  
(Given: acceleration due to gravity on the surface of earth is  $g$ )

(A)  $mgR$

(B)  $\frac{mgR}{2}$

(C)  $\frac{mgR}{4}$

(D)  $\frac{3mgR}{4}$

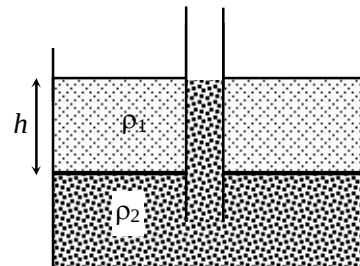
Ans. C

Sol. Conserving momentum during the explosion  $mv = \frac{m}{2} \times 0 + \frac{m}{2} v'$  or  $v' = 2v$

$$\text{Increase in the mechanical energy} = \Delta K + \Delta U = \Delta K + 0 = \frac{1}{2} \frac{m}{2} (2v)^2 - \frac{1}{2} mv^2 = \frac{1}{2} mv^2$$

$$= \frac{GMm}{4R} = \frac{mgR}{4} \quad \left[ v = \sqrt{\frac{GM}{2R}} \right]$$

8. A container has two immiscible liquids of densities  $\rho_1$  and  $\rho_2$  ( $\rho_2 > \rho_1$ ). A capillary tube of radius  $r$  is inserted in the liquid so that its bottom reaches upto the denser liquid. The denser liquid rises in the capillary and attains a height  $h$  from the interface of the liquids, which is equal to the column length of the lighter liquid. Assuming angle of contact to be zero, the surface tension of heavier liquid is



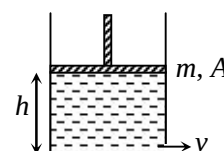
(A)  $2\pi r \rho_2 g h$

- (B)  $\frac{\rho_2 r g h}{2}$
- (C)  $\frac{r}{2}(\rho_2 - \rho_1) g h$
- (D)  $2\pi r(\rho_2 - \rho_1) g h$

Ans. C

Sol.  $P_0 + h\rho_1 g - \rho_2 g h + \frac{2T}{r} = P_0, \quad T = \frac{r}{2}(\rho_2 - \rho_1) g h$

9. A cylindrical vessel contains a liquid of density  $\rho$  upto a height  $h$ . The liquid is closed by a piston of mass  $m$  and area of cross-section  $A$ . There is a small hole at the bottom of the vessel. The speed  $v$  with which the liquid comes out of the hole is: (neglect presence of atmosphere)



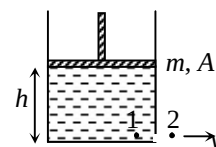
- (A)  $\sqrt{2gh}$
- (B)  $\sqrt{2\left(gh + \frac{mg}{\rho A}\right)}$
- (C)  $\sqrt{2\left(gh + \frac{mg}{A}\right)}$
- (D)  $\sqrt{2gh + \frac{mg}{A}}$

Ans. B

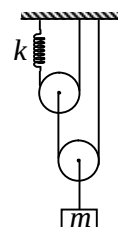
Sol. Applying Bernoulli's theorem at 1 and 2

$$\rho g h + \frac{mg}{A} = \frac{1}{2} \rho v^2$$

or  $v = \sqrt{2\left(gh + \frac{mg}{\rho A}\right)}$



10. Find the natural frequency of the system as shown in figure. The pulleys are massless and frictionless.



(A)  $\frac{2}{\pi} \sqrt{\frac{k}{m}}$

(B)  $\frac{1}{2\pi} \sqrt{\frac{k}{m}}$

(C)  $\frac{4}{\pi} \sqrt{\frac{k}{m}}$

(D)  $\frac{8}{\pi} \sqrt{\frac{k}{m}}$

Ans. A

Sol. Let  $m$  is displaced downward by  $x$  from its mean position then spring will stretched by  $x_0 = 4x$ .

$$\Rightarrow x = \frac{x_0}{4}, \quad m\omega^2 x = -4kx_0 = -16kx \quad \Rightarrow f = \frac{2}{\pi} \sqrt{\frac{k}{m}}$$

11. A stationary wave is set up on a string fixed at both ends. The distance between two consecutive nodes is 18 cm at a particular mode of vibration and for the next higher mode of vibration in the same string the distance between two consecutive nodes is 16 cm. The length of string is

(A) 144 cm

(B) 140 cm

(C) 36 cm

(D) 32 cm

Ans. A

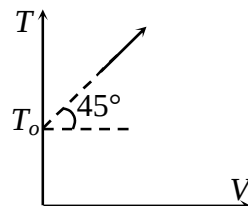
Sol.  $n \frac{\lambda_1}{2} = (n+1) \frac{\lambda_2}{2} \quad \dots (i)$

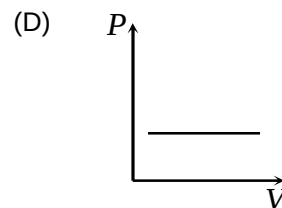
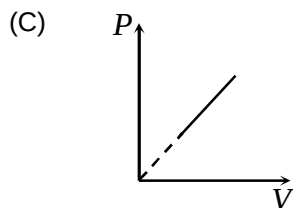
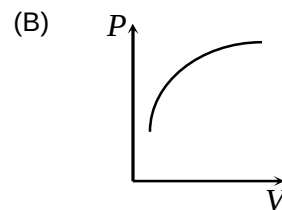
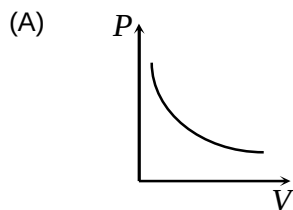
(Given  $\lambda_1 = 36, \lambda_2 = 32$ )

$\therefore n = 8$

$\frac{n\lambda_1}{2} = l \quad \therefore l = 144 \text{ cm}$

12. The give curve represents the variation of temperature as a function of volume for one mole of an ideal gas. Which of the following curves represents the variation of pressure as a function of volume?





Ans. A

Sol.  $T = V \tan 45^\circ + T_0$   
 $T = V + T_0$

For one mole of an ideal gas  $T = \frac{PV}{R}$

$$\therefore \frac{PV}{R} = V + T_0 \quad \therefore (P - R)V = RT_0$$

Hence graph is rectangular hyperbola given in option (a).

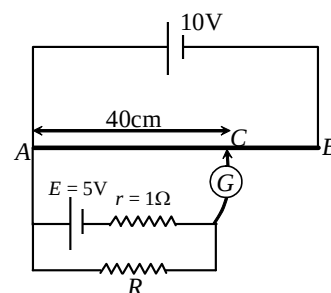
13. A solid conducting sphere having a charge  $Q$  is surrounded by an uncharged concentric conducting hollow spherical shell. The potential difference between the surface of solid sphere and the outer surface of the hollow shell is  $V$ . If the shell is now given a charge of  $-3Q$ , the new potential difference between the same two surfaces is

- (A)  $V$   
 (B)  $2V$   
 (C)  $4V$   
 (D)  $-2V$

Ans. A

Sol. In the given situation potential difference is independent of charge on outer shell.

14. A potentiometer wire  $AB$  is 100cm long and has total resistance of  $10\Omega$ . Find the value of unknown resistance  $R$  so that null point is obtained at a distance 40 cm from  $A$ .

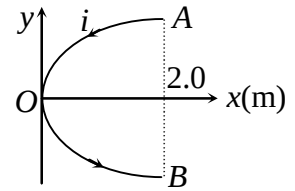


- (A)  $1\ \Omega$   
 (B)  $2\ \Omega$   
 (C)  $3\ \Omega$   
 (D)  $4\ \Omega$

Ans. D

Sol.  $\frac{10}{100} \times 40 = 5 - \frac{5}{R+1} \times 1 \Rightarrow R = 4\ \Omega$

15. A conducting wire bent in the form of a parabola  $y^2 = 2x$  carries a current  $i = 2\text{A}$  as shown in figure. This wire is placed in a uniform magnetic field  $\vec{B} = -4\hat{k}$  tesla. The magnetic force on the wire is

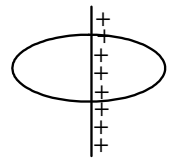


- (A)  $-16\hat{i}$   
 (B)  $32\hat{i}$   
 (C)  $-32\hat{i}$   
 (D)  $16\hat{i}$

Ans. B

Sol.  $\vec{F}_m = 2 \left[ 4(-\hat{j}) \times 4(-\hat{k}) \right] = \vec{F}_m = 32\hat{i}$

16. A very long uniformly charged rod falls with a constant velocity  $V$  through the center of a circular loop. Then the magnitude of induced emf in loop is (charge per unit length of rod =  $\lambda$ )



- (A)  $\frac{\mu_0}{2\pi} \lambda V^2$   
 (B)  $\frac{\mu_0}{2} \lambda V^2$   
 (C)  $\frac{\mu_0}{2\lambda} V$   
 (D) zero

Ans. D

Sol. Magnetic flux linked with the loop is always zero.  
So induced emf is also zero

17. A capacitor of capacity  $2 \mu\text{F}$  is charged to a potential difference of  $12\text{V}$ . It is then connected across an inductor of inductance  $0.6 \text{ mH}$ . The current in the circuit at a time when the P.D. across the capacitor is  $6.0 \text{ V}$  is

- (A)  $0.6 \text{ A}$   
(B)  $1.2 \text{ A}$   
(C)  $2.4 \text{ A}$   
(D)  $3.6 \text{ A}$

Ans. A

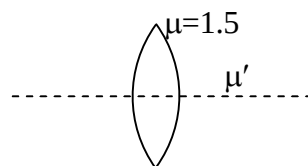
Sol.  $\omega = \frac{1}{\sqrt{LC}} = \frac{10^5}{\sqrt{12}}$   
 $q = q_0 \cos \omega t$   
 $2 \times 10^{-6} \times 6 = 2 \times 10^{-6} \times 12 \cos \omega t$   
 $\cos \omega t = \frac{1}{2} \Rightarrow \omega t = \frac{\pi}{3}$   
 $i = \frac{dq}{dt} = -q_0 \omega \sin \omega t$   
 $i = 6 \text{ A}$

18. A thin converging lens of refractive index  $1.5$  has power of  $+5\text{D}$ . When this lens is immersed in a liquid, it acts as a diverging lens of focal length  $100 \text{ cm}$ . The refractive index of the liquid is

- (A)  $2$   
(B)  $\frac{15}{11}$   
(C)  $\frac{4}{3}$   
(D)  $\frac{5}{3}$

Ans. D

Sol.  $P = \frac{1}{f} = (\mu - 1) \frac{2}{R}$   
 $P = (1.5 - 1) \frac{2}{R} = 5$



$$R = 20 \text{ cm}$$

Focal length of this lens in medium ( $\mu'$ )

$$\frac{1}{f'} = \left( \frac{\mu}{\mu'} - 1 \right) \frac{2}{R} = \left( \frac{1.5}{\mu'} - 1 \right) \frac{2}{20}$$

$$\frac{-1}{10} = \left( \frac{1.5}{\mu'} - 1 \right) \frac{1}{10} \Rightarrow \frac{1.5}{\mu'} = 1 - \frac{1}{10} = \frac{9}{10}$$

$$\mu' = \frac{15}{9} = \frac{5}{3}$$

19. Unpolarized light of intensity  $32 \text{ Wm}^{-2}$  passes through three polarizers such that transmission axes of the first and second polarizer makes an angle  $30^\circ$  with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of final emerging light will be

- (A)  $32 \text{ Wm}^{-2}$   
 (B)  $3 \text{ Wm}^{-2}$   
 (C)  $8 \text{ Wm}^{-2}$   
 (D)  $4 \text{ Wm}^{-2}$

Ans. B

Sol.  $I_0 = 32 \text{ Wm}^{-2}$

After passing through first polarizer,  $I_1 = \frac{I_0}{2} = 16 \text{ W/m}^2$

After passing through second polarizer,  $I_2 = I_1 \cos^2 30^\circ = 12 \text{ W/m}^2$

After passing through third polarizer,  $I_3 = I_2 \cos^2 60^\circ = 3 \text{ W/m}^2$

20. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy  $T_A$  eV and de Broglie wavelength  $\lambda_A$ . The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is  $T_B = (T_A - 1.50) \text{ eV}$ . If the de Broglie wavelength of these photoelectrons is  $\lambda_B = 2\lambda_A$ , then choose the incorrect statement.

- (A) the work function of A is 1.25 eV  
 (B) the work function of B is 4.20 eV  
 (C)  $T_A = 2.00 \text{ eV}$   
 (D)  $T_B = 0.5 \text{ eV}$

Ans. A

Sol.  $\lambda_B = 2\lambda_A \Rightarrow 4T_B = T_A$

$$T_B = T_A - 1.5$$

$$T_B = 0.5 \text{ eV}$$

$$T_A = 2 \text{ eV}$$

$$\therefore \phi_A = 2.25 \text{ eV and } \phi_B = 4.2 \text{ eV}$$

### SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A binary star consists of two stars A (mass  $2.2 M_S$ ) and B (mass  $11 M_S$ ), where  $M_S$  is the mass of the sun. They are separated by distance  $d$  and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is

Ans. 00006.00

Sol. 
$$\frac{L_{\text{total}}}{L_B} = \frac{(I_A + I_B)\omega}{I_B\omega} = \frac{m_A r_A^2}{m_B r_B^2} + 1 = \frac{11}{2.2} + 1 = 6$$

22. Degree of freedom for a gas is 6. This gas does 25 J of piston work at constant pressure. Net heat absorbed by the gas during this process is .....

Ans. 00075.00

Sol. Work done at constant pressure is

$$\Delta W = \left(1 - \frac{1}{\gamma}\right) \times \Delta Q_{\text{total}}$$

$$\Rightarrow 25 = \left(1 - \frac{1}{\gamma}\right) \Delta Q_{\text{total}}$$

But,  $\gamma = 1 + \frac{2}{f} = 1 + \frac{2}{6}$

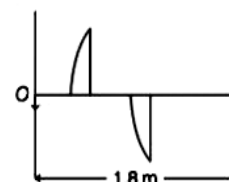
or  $\gamma = \frac{4}{3}$

So,  $25 = \left(1 - \frac{3}{4}\right) \times \Delta Q_{\text{total}}$

or  $\Delta Q_{\text{total}} = 100 \text{ J}$

So, net heat absorbed by the gas =  $100 - 25 = 75 \text{ J}$

23. A thin plano-convex lens of focal length  $f$  is split into two halves. One of the halves is shifted along the optical axis. The separation between object and image planes is 1.8 m. The magnification of the image formed by one of the half lens is 2. Find the separation between the halves.



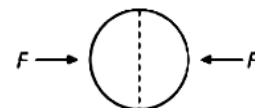
Ans. 00000.60

Sol.  $\left| \frac{v}{u} \right| = 2$  or  $|u| = 2|u|$

When  $u = -x$ , then  $v = +2x$  and  $|x| + |v| = 1.8\text{m}$

$$\Rightarrow 3x = 1.8\text{m} \text{ or } x = 0.60\text{m}$$

24. A uniformly charged thin spherical shell of radius  $R$  carries uniform surface charge density  $\sigma$  per unit area. It is made of two hemispherical shells, held together by pressing them with force  $F$ . The value of  $F$  will be (Take,  $\pi\sigma^2 R^2 = 4\epsilon_0$ )



Ans. 00002.00

Sol. For the given area total force will be  $u \, dA$   
When  $dA$  is the area, then electrical force per unit area

$$= \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \epsilon_0 \left( \frac{\sigma}{2\epsilon_0} \right)^2 = \frac{\sigma^2}{2g\epsilon_0}$$

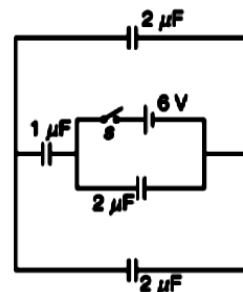
$$\text{Projection area} = \pi r^2$$

$$\Rightarrow \text{Net electrical force} = \left( \frac{\sigma^2}{2\epsilon_0} \right) (\pi R^2)$$

In equilibrium, this force should be equal to the applied force,

$$F = \frac{\sigma^2 R^2}{2\epsilon_0} = \frac{4\epsilon_0}{2\epsilon_0} = 2$$

25. Find charge supplied by the cell after the switch is closed (in  $\mu\text{C}$ )

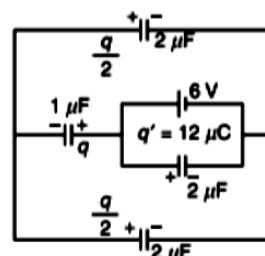


Ans. 00016.80

Sol. The potential difference across  $1 \mu\text{F}$  and  $2 \mu\text{F}$  (uppermost) capacitor must sum up to

$$\frac{q}{1} + \frac{q/2}{2} = 6 \Rightarrow q = \frac{24}{5} \mu\text{C}$$

$$\therefore \text{Answer is } 12 + \frac{24}{5} = 16.8 \mu\text{C}$$



26. Light of wavelength  $1800 \text{ \AA}$  ejects photoelectrons from a plane of a metal whose work function is  $2 \text{ eV}$ . If a uniform magnetic field of  $5 \times 10^{-5} \text{ tesla}$  is applied parallel to plate, what would be the radius of the path (in metre) followed by electrons ejected normally from the plate with maximum energy.

Ans. 00000.15

Sol. Energy of incident photons in eV is given as

$$E = \frac{12431}{1800} \text{ eV}$$

As work function of metal is 2eV, the maximum kinetic energy of ejected electrons is

$$KE_{\max} = E - \phi$$

$$\Rightarrow KE_{\max} = 6.9 - 2 \text{ eV}$$

$$\Rightarrow KE_{\max} = 4.9 \text{ eV}$$

If  $v_{\max}$  be the speed of fastest electrons then we have

$$\frac{1}{2} m v_{\max}^2 = 4.9 \times 1.6 \times 10^{-19} \text{ joule}$$

$$\Rightarrow v_{\max} = \sqrt{\frac{2 \times 4.9 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}}$$

$$\Rightarrow v_{\max} = 1.31 \times 10^6 \text{ m/s}$$

When an electron with this speed enters a uniform magnetic field normally it follows a circular path whose radius can be given by

$$r = \frac{mv}{qB} \quad [\text{As } qvB = \frac{mv^2}{r}]$$

$$\Rightarrow r = \frac{9.1 \times 10^{-31} \times 1.31 \times 10^6}{1.6 \times 10^{-19} \times 5 \times 10^{-5}}$$

$$\Rightarrow r = 0.149 \text{ m}$$

27. A U tube contains water and methylated spirit separated by mercury. The mercury columns in the two arms are at the same level with 10.0 cm of water in one arm and 12.5 cm of spirit in the other. What is the relative density of spirit?

Ans. 00000.80

Sol. For water column in one arm of U tube,  $h_1 = 10.0 \text{ cm}$ ;  $\rho_1$  (density =  $1 \text{ g cm}^{-3}$ )

For spirit column in other arm of U tube,  $h_2 = 12.5 \text{ cm}$ ;  $\rho_2 = ?$

As the mercury columns in the two arms of U tube are in level, therefore pressure exerted by each is equal.

$$\text{Hence } h_1 \rho_1 g = h_2 \rho_2 g \text{ or } \rho_2 = \frac{h_1 \rho_1}{h_2} = \frac{10 \times 1}{12.5} = 0.8 \text{ g cm}^{-3}$$

Therefore, relative density of spirit =  $\rho_2 / \rho_1 = 0.8 / 1 = 0.8$

28. In an  $\alpha$ -decay the kinetic energy of  $\alpha$ -particle is 48 MeV and Q-value of the reaction is 50 MeV. The mass number of the mother nucleus is X. Find the value of X/4. (Assume that daughter nucleus is in ground state)

Ans. 00025.00

Sol. We have  $K_\alpha = \frac{m_y}{m_y + m_\alpha} \cdot Q$

$$\Rightarrow K_\alpha = \frac{A-4}{A} \cdot Q \Rightarrow 48 = \frac{A-4}{A} \cdot 50 \Rightarrow A = 100$$

29. Electrons in hydrogen like atom ( $Z = 3$ ) make transitions from the fifth to the fourth orbit and from the fourth to the third orbit. The resulting radiations are incident normally on a metal plate and eject photoelectrons. The stopping potential for the photoelectrons ejected by the shorter wavelength is 3.95 volts. Calculate the work function of the metal (in eV). (Rydberg constant =  $1.094 \times 10^7 \text{ m}^{-1}$ )

Ans. 00002.00

Sol. For hydrogen like atom energy of the  $n$ th orbit is

$$E_n = -\frac{13.6}{n^2} Z^2 \text{ eV/atom}$$

For transition from  $n = 5$  to  $n = 4$ ,

$$h\nu = 13.6 \times 9 \left[ \frac{1}{16} - \frac{1}{25} \right] = \frac{13.6 \times 9 \times 9}{16 \times 25} = 2.754 \text{ eV}$$

For transition from  $n = 4$  to  $n = 3$ ,

$$h\nu' = 13.6 \times 9 \left[ \frac{1}{9} - \frac{1}{16} \right] = \frac{13.6 \times 9 \times 7}{9 \times 16} = 5.95 \text{ eV}$$

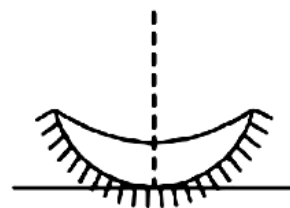
For transition  $n = 4$  to  $n = 3$ , the frequency is high and hence wavelength is short.

For photoelectric effect,  $h\nu' - W = eV_0$ , where  $W$  = work function

$$5.95 \times 1.6 \times 10^{-19} - W = 1.6 \times 10^{-19} \times 3.95$$

$$\Rightarrow W = 2 \times 1.6 \times 10^{-19} = 2 \text{ eV}$$

30. The convex surface of a thin concavo-convex lens of glass of refractive index 1.5 has a radius of curvature 20 cm. The concave surface has a radius of curvature 60 cm. The convex side is silvered and placed on a horizontal surface. At what distance should a pin be placed on optical axis so that image is formed at same place.



Ans. 00015.00

Sol. This silvered concavo-convex lens behaves like a mirror whose focal length can be calculated by the formula

$$\frac{1}{f} = \frac{2}{f_1} + \frac{1}{f_2}$$

$f_1$  = focal length of concave surface.

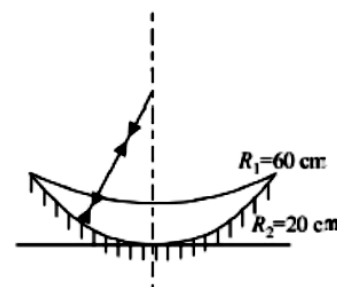
$f_2$  = focal length of concave mirror.

$$\therefore \frac{1}{f} = \frac{2}{-30} + \frac{1}{-10} = \frac{4}{30}$$

$$\therefore f = -7.5 \text{ cm}$$

Using mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{-7.5} = \frac{1}{-x} + \frac{1}{-x} \Rightarrow x = 15 \text{ cm}$$



# Chemistry

## PART – B

### SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. When the letters shown have their usual meanings, the velocity of photoelectrons is:

(A)  $\left[ \frac{2hc}{m} \left\{ \frac{\lambda_0 - \lambda}{\lambda \lambda_0} \right\} \right]^{1/2}$

(B)  $\left[ \frac{2hc}{m} \left\{ \frac{\lambda_0 - \lambda}{\lambda \lambda_0} \right\} \right]^2$

(C)  $\frac{2hc}{m}$

(D)  $\left( \frac{m}{2hc} \right)^2$

Ans. A

Sol.  $KE = h(\nu - \nu_0)$

$$\frac{1}{2}mv^2 = hc \left( \frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

$$v^2 = \frac{2hc}{m} \left( \frac{\lambda_0 - \lambda}{\lambda \lambda_0} \right)$$

32. Which of the following properties is not shown by NO?

- (A) It is a neutral oxide
- (B) It combines with oxygen to form nitrogen dioxide
- (C) Its bond order is 2.5
- (D) It is diamagnetic in gaseous state

Ans. D

Sol. NO has total 15 electrons and one electron is unpaired.

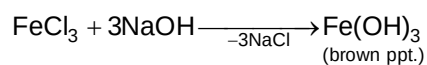
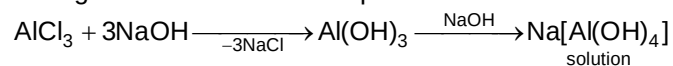
33.  $AlCl_3$  and  $FeCl_3$  can be separated from their mixture by using:

- (A)  $NH_4OH$
- (B)  $NaOH$

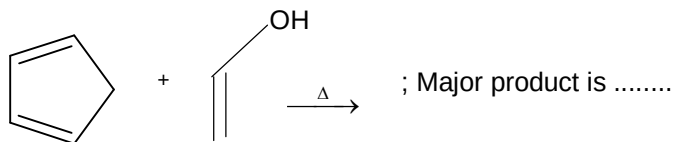
- (C)  $\text{H}_2\text{O}$   
 (D) magnetic method

Ans. B

Sol.  $\text{AlCl}_3$  dissolves in aqueous  $\text{NaOH}$  while  $\text{FeCl}_3$  gives brown ppt. of  $\text{Fe}(\text{OH})_3$  which on separation can again be converted to respective chlorides.



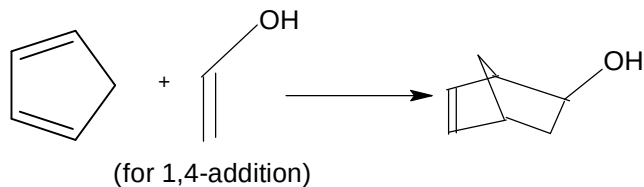
34.



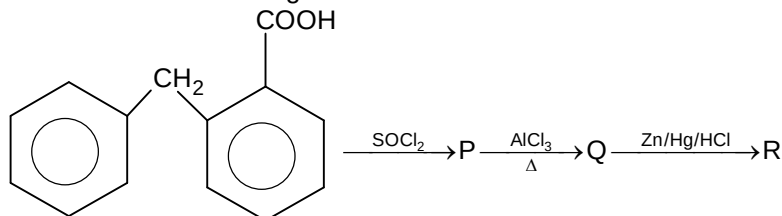
- (A)
- (B)
- (C)
- (D)

Ans. A

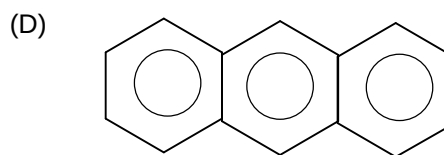
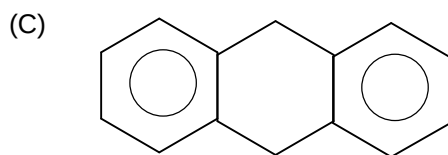
Sol.



35. What is R in the following set of reactions?

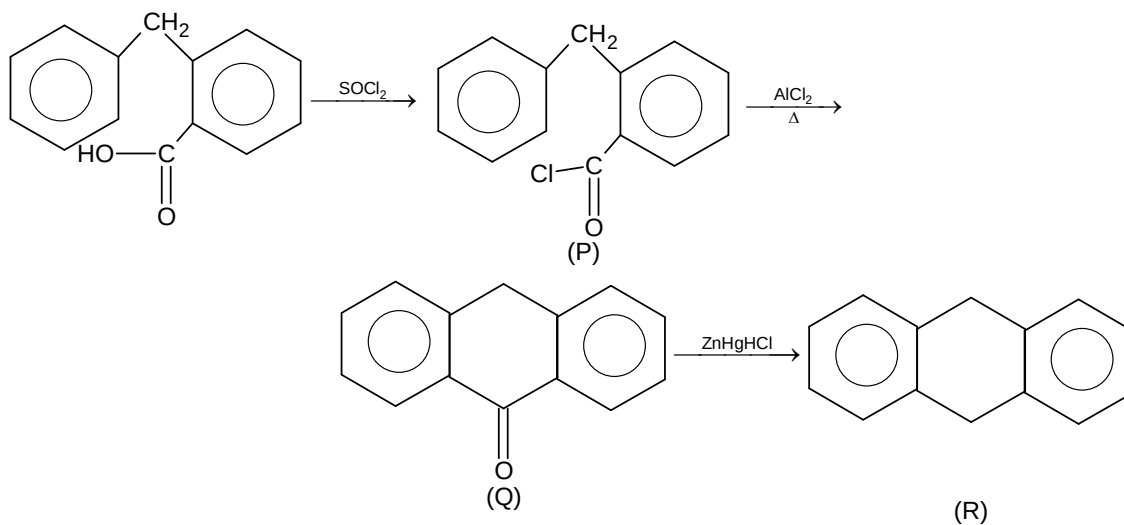


- (A)
- (B)



Ans. C

Sol.



36. Which of the following species has the shortest bond length?

- (A)  $\text{NO}^{2-}$
- (B)  $\text{NO}^+$
- (C)  $\text{NO}$
- (D)  $\text{NO}^-$

Ans. B

Sol. MOT of NO,

$$\sigma_{1s}^2 < \sigma_{1s}^{*2} < \sigma_{2s}^2 < \sigma_{2s}^{*2} < \pi_{2px}^2 = \pi_{2py}^2 < \sigma_{2pz}^2 < \pi_{2px}^{*1} = \pi_{2py}^*$$

$$\text{B.O.} = \frac{10 - 5}{2} = 2.5$$

For  $\text{NO}^+$

$$\text{B.O.} = \frac{10 - 4}{2} = 3$$

For  $\text{NO}^{2-}$

$$\text{B.O.} = \frac{10 - 7}{2} = 1.5$$

$$\text{For } \text{NO}^-, \text{B.O.} = \frac{10 - 6}{2} = 2$$

$$\text{Bond length} \propto \frac{1}{\text{Bond order}}$$

$$\text{SO}_4^{2-} \xrightarrow{-0.936} \text{SO}_3^{2-} \xrightarrow{-0.576} \frac{1}{2} \text{S}_2\text{O}_3^{2-}$$

$\text{E}^0 = ?$

- (A) 0.326 V  
(B) 0.425 V  
(C) 0.756 V  
(D) 0.512

Sol.

$$\Delta G^{\circ} = G_1^{\circ} + \Delta G_2^{\circ}$$
$$-nFE^{\circ} = -nFE_1^{\circ} - nFE_2^{\circ}$$
$$-4FE^{\circ} = -2F \times (-0.936) - 2F(0.576)$$
$$-4E^{\circ} = 2 \times 0.936 + 2 \times 0.576$$
$$E^{\circ} = \frac{1.872 + 1.152}{4} = 0.756$$

(A) pH = 8  
(B) pH = 9  
(C) pH = 8.5  
(D) pH = 9.5

Sol.

(1)  $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

(2)  $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

$C = \frac{20}{200} = 0.1\text{M}$

$\text{pH} = 7 + \frac{1}{2}\{5 + \log(0.1)\} = 9$

(A)  $\text{PF}_5$

(B)  $\text{SF}_4$

(C)  $\text{BiI}_5$

(D)  $\text{BiI}_3$

Ans. C

Sol.  $\text{BiI}_5$  doesn't exist due to the inert pair effect of Bi and reducing property of  $\text{I}^-$ .

40. For equilibrium process  $\text{S}_8(\text{g}) \rightleftharpoons 4\text{S}_2(\text{g})$  reaction starts with  $\text{S}_8(\text{g})$ , at equilibrium V.D. of mixture is found to be 64. Calculate degree of dissociation of  $\text{S}_8$ .

(A)  $2/3$

(B)  $1/3$

(C)  $4/3$

(D)  $5/3$

Ans. B

Sol.  $\text{S}_8 \rightleftharpoons 4\text{S}_2$   
 $\begin{matrix} 1-\alpha & & 4\alpha \end{matrix}$   
 $1 + 3\alpha = \frac{32 \times 8}{32 \times 4} = 2$   
 $\Rightarrow \alpha = \frac{1}{3}$

41. What is the free energy change for the process:



(s,  $-10^\circ\text{C}, 0.26 \text{ Pa}, 1 \text{ mole}$ )

Given that vapour pressure of water and ice at  $-10^\circ\text{C}$  is 0.28 and 0.26 Pa respectively.

(A)  $R \times 263 \times \ln \frac{14}{13}$

(B)  $R \times 263 \times \ln \frac{13}{14}$

(C)  $-R \times 10 \times \ln \frac{13}{14}$

(D) Zero

Ans. B

Sol.  $dG = VdP - SdT$   
 $dG = VdP$  (T=constant)  
 $\Delta G = nRT \ln \left( \frac{P_2}{P_1} \right) = 263R \ln \left( \frac{0.26}{0.28} \right)$

$$= 263R \ln \left( \frac{13}{14} \right)$$

42. Beryllium shows diagonal relationship with aluminium, which of the following is incorrect?

- (A)  $\text{Be}_2\text{C}$  like  $\text{Al}_4\text{C}_3$  yields methane on hydrolysis
- (B) Be, like Al is not rendered passive by  $\text{HNO}_3$
- (C)  $\text{Be}(\text{OH})_2$  like  $\text{Al}(\text{OH})_3$  is basic
- (D) Be forms beryllates and Al forms aluminates

Ans. C

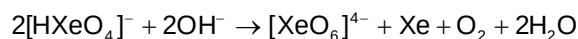
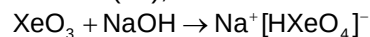
Sol. Factual

43. Select the incorrect statement.

- (A) In alkaline solution  $\text{XeO}_3$  forms xenate ions
- (B) Perxenate ions can be formed by disproportionation of xenates in alkaline conditions
- (C) In perxenates Xe is in +6 oxidation state
- (D) Perxenates can act as oxidizing agents

Ans. C

Sol. Xenates (+6), Persenates (+8)



44. Which one of the following pairs of isomers and types of isomerism are correctly matched-

- (P)  $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$  and  $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2$ ... linkage
- (Q)  $[\text{Cu}(\text{NH}_3)_4](\text{PtCl}_4)$  and  $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$ .... co-ordination
- (R)  $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$  and  $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$ ....ionization

Select the correct answer using the codes given below:

- (A) Q and R
- (B) P, Q and R
- (C) P and R
- (D) P and Q

Ans. B

Sol. Factual

45. IUPAC name of the complex compound  $\text{Na}_3[\text{Cr}(\text{O})_2(\text{O}_2)(\text{O}_2)_2(\text{NH}_3)]$  where half of the coordination number are occupied by neutral ligands with magnetic moment  $\mu = \sqrt{15}$  must be:

- (A) Sodium amminedioxodioxysuperoxochromate (III)
- (B) Sodium amminedioxygendioxodisuperoxochromate (III)
- (C) Sodium amminebisdioxygendioxoperoxochromate (III)
- (D) Sodium amminebisdioxygendioxoperoxochromium (III)

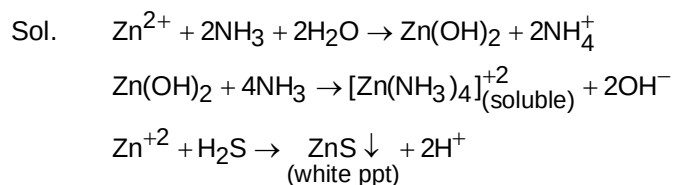
Ans. C

Sol. Factual

46. An aqueous solution of colourless metal sulphate M, gives a white ppt., with  $\text{NH}_4\text{OH}$ . This was soluble in excess of  $\text{NH}_4\text{OH}$ . On passing  $\text{H}_2\text{S}$  through this solution a white ppt. is formed. The metal M in the salt is:

- (A) Ca
- (B) Ba
- (C) Al
- (D) Zn

Ans. D



47. A pale green crystalline metal salt of M dissolves freely in water. On standing it gives a brown ppt. On addition of aqueous  $\text{NaOH}$ . The metal salt solution also gives a black ppt. On bubbling  $\text{H}_2\text{S}$  in basic medium. An aqueous solution of the metal salt decolourizes the pink colour of the permanganate solution. The metal in the metal salt solution is:

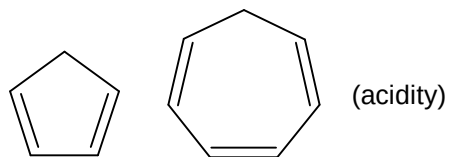
- (A) copper
- (B) aluminium
- (C) lead
- (D) iron

Ans. D

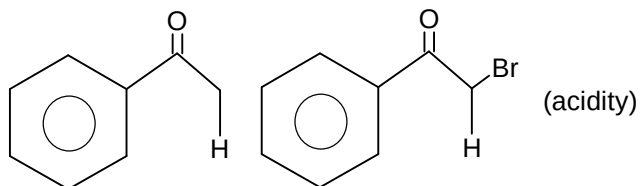
Sol. The salt could be iron (II) sulphate, in the presence of  $\text{NaOH}$  and air is oxidised to ferric hydroxide. On passing  $\text{H}_2\text{S}$  forms  $\text{FeS}$  which is black in colour, potassium permanganate oxidises  $\text{Fe}^{+2}$  to  $\text{Fe}^{3+}$  and itself get reduced to  $\text{Mn}^{+2}$  due to which the pink colour disappears.

48. In which of the following cases the compound written on right shows the mentioned phenomenon more than the compound written on left?

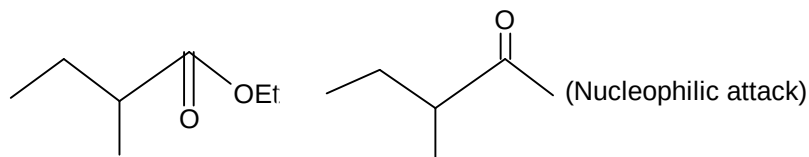
(p)



(q)



(r)



(A) p, q

(B) q, r

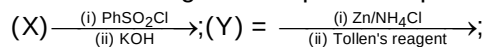
(C) p, r

(D) in all

Ans. B

Sol. Factual.

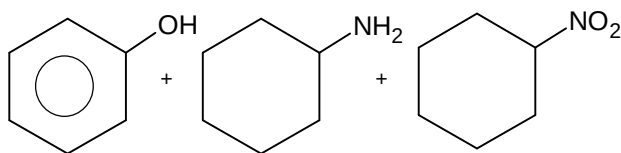
49. A mixture of organic compound responded positively for following reagents (X, Y, Z):



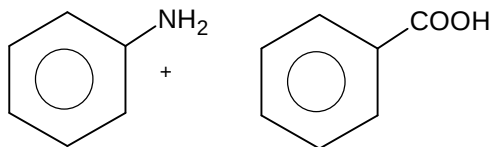
(Z) = Na metal.

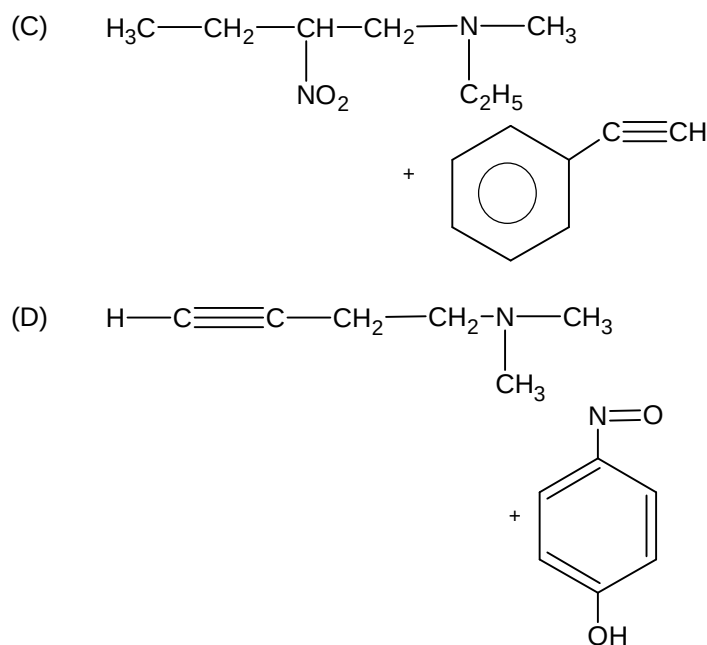
The mixture has:

(A)



(B)





Ans. A

Sol. A primary amine gives positive test with Hinsberg reagent ( $\text{PhSO}_2\text{Cl}$ ), a  $-\text{NO}_2$  group gives positive test with mulliken's reagent ( $\text{Zn}/\text{NH}_4\text{Cl}$ , Tollen's reagent and an acidic functional group gives positive test with sodium metal.

50. In Fishcher-Killiani synthesis, the anomeric carbon of the reactant:

- (A) remains as carbonyl
- (B) is lost as  $\text{CO}_2$
- (C) becomes the new epimeric centre
- (D) becomes a nitrile carbon.

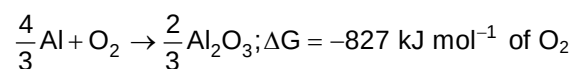
Ans. C

Sol. Fischer-Killiani synthesis produces C-2 epimers.

### SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. On the basis of the information available from the reaction;



The minimum EMF required to carry out electrolysis of  $\text{Al}_2\text{O}_3$  is:  
(1 F = 96500) C  $\text{mol}^{-1}$  of electrons)

Ans. 00002.14

Sol. 1 mol Al,  $n = 3$  (electron exchanged)

$$\frac{4}{3} \text{ mol Al}, n = \frac{4}{3} \times 3 = 4$$

$$\Delta G^0 = -nFE^0 \Rightarrow E^0 = 2.14V$$

52. Calculate  $\gamma$  (ratio of  $C_p$  and  $C_v$ ) for triatomic linear gas at high temperature. Assume that the contribution of vibrational degree of freedom is 75%:

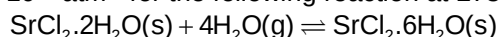
Ans. 00001.18

Sol.  $C_v = \frac{3}{2}R + R + 4R \times \frac{3}{4} = \frac{11}{2}R;$

$$C_p = \frac{11}{2}R + R = \frac{13}{2}R$$

$$\frac{C_p}{C_v} = \frac{13}{11} = 1.18$$

53. In the presence of excess of anhydrous  $\text{SrCl}_2$ , the amount of water taken up is governed by  $K_p = 10^{12} \text{ atm}^{-4}$  for the following reaction at 273 K



What is equilibrium vapour pressure (in torr) of water in a closed vessel that contains  $\text{SrCl}_2 \cdot 2\text{H}_2\text{O}(g)$ ?

Ans. 00000.76

Sol.  $K_p = \frac{1}{P_{\text{H}_2\text{O}(g)}^4}$

$$\Rightarrow P_{\text{H}_2\text{O}(g)} = \left( \frac{1}{K_p} \right)^{1/4}$$

$$\Rightarrow (10^{-12})^{1/4} = 10^{-3} \text{ atm}$$

$$\therefore K_p = 10^{-3} \times 760 = 0.76 \text{ torr}$$

54. Consider the following first order competing reactions:



If 50% of the reaction of X was completed when 96% of the reaction of Y was completed, the ratio of their rate constants ( $k_2/k_1$ ) is:

Ans. 00004.65

Sol.  $K_2 = \frac{1}{t} \ln \frac{100}{4} \quad \dots(i)$

$$K_1 = \frac{1}{t} \ln \frac{100}{50} \quad \dots(ii)$$

$$\frac{K_2}{K_1} = \frac{\ln 25}{\ln 2} = 4.65$$

55. If 0.50 L of a 0.60 M  $\text{SnSO}_4$  solution is electrolysed for a period of 30.0 min using a current of 4.60 A. If inert electrodes are used, what is the final concentration of  $\text{Sn}^{2+}$  remaining in the solution?  
[At. Mass of Sn = 119]

Ans. 00000.51

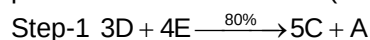
Sol.  $\frac{W}{M} \times 2 = \frac{4.6 \times 30 \times 60}{96500}$   
Moles of  $\text{Sn}^{2+}$  reduced = 0.043  
Initial moles of  $\text{Sn}^{2+} = 0.5 \times 0.6 \Rightarrow 0.30$ ;  
Remaining moles of  $\text{Sn}^{2+} = 0.30 - 0.043$   
 $\therefore [\text{Sn}^{2+}] = \frac{0.257}{0.5} = 0.514\text{M}$

56. 9.2 grams of  $\text{N}_2\text{O}_4(\text{g})$  is taken in a closed one litre vessel and heated till the following equilibrium is reached  $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$   
At equilibrium, 50%  $\text{N}_2\text{O}_4(\text{g})$  is dissociated. What is the equilibrium constant (in  $\text{mol litre}^{-1}$ )  
(molecular mass of  $\text{N}_2\text{O}_4 = 92$ )

Ans. 00000.20

Sol.  $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$   
 $t = 0 \quad 0.0(1-\alpha) \quad 2(0.1\alpha)$   
 $\therefore \alpha = 0.5$   
 $K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]}$

57. 9 moles of "D" and 14 moles of E are allowed to react in a closed vessel according to given reactions. Calculate number of moles of B formed in the end of reaction, if 4 moles of G are present in reaction vessel. (Percentage yield of reaction is mentioned in the reaction)



Ans. 00002.40

Sol. L.R.  
 $3\text{D} + 4\text{E} \xrightarrow{80\%} 5\text{C} + \text{A}$   
9 mole 14 mole  $\frac{5}{3} \times 9 \times 0.8 = 12$  mole  
 $3\text{C} + 5\text{G} \xrightarrow{50\%} 6\text{B} + \text{F}$   
12 mole 4 mole  
L.R. is G  
Moles of B formed  $= \frac{6}{5} \times 4 \times 0.5 = 2.4$

58. A sample of 28 mL of  $\text{H}_2\text{O}_2$  (aq) solution required 10 mL of 0.1M  $\text{KMnO}_4$  (aq) solution for complete reaction in acidic medium. What is the volume strength of  $\text{H}_2\text{O}_2$ ?

Ans. 00001.00

Sol. Eq. of  $\text{H}_2\text{O}_2 = \text{Eq. of KMnO}_4$   
 $(25 \times M \times 2) = (10 \times 0.1 \times 5)$   
 $M = 0.089$   
 Volume strength =  $M \times 0.1 = 1$

59. Calculate the value of A.

$A = \frac{E_{1,2}}{2E_{2,1}}$  where  $E_{n,Z}$  = Energy of electron in nth orbit, Z = atomic number of hydrogen like species.

Ans. 00008.00

Sol.  $A = \frac{E_{1,2}}{2E_{2,1}} = \frac{-13.6 \times 2^2 \times 2^2}{2 \times 1^2 \times (-13.6) \times 1^2} = \frac{16}{2} = 8$

60. 2 mole of zinc is dissolved in HCl at  $25^\circ\text{C}$ . the magnitude of work done in open vessel is (in kJ)

Ans. 00004.95

Sol.  $w = -P \cdot \Delta V = -n_{\text{H}_2} RT$   
 $RT = -\frac{2 \times 8.314 \times 298}{1000} = -4.955 \text{ kJ}$

# Mathematics

## PART – C

### SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The vectors  $\vec{a}$  and  $\vec{b}$  are non-zero and non-collinear. The value of  $x$  for which vector  $\vec{c} = (x-2)\vec{a} + \vec{b}$  and  $\vec{d} = (2x+1)\vec{a} - \vec{b}$  are collinear.

- (A) 1  
(B)  $\frac{1}{2}$   
(C)  $\frac{1}{3}$   
(D) 2

Ans. C

Sol. Since  $\vec{c}$  and  $\vec{d}$  are collinear.

We have  $p\vec{c} + q\vec{d} = 0$  where  $p + q = 0$  for some scalars  $p$  and  $q$ .

$$p\{(x-2)\vec{a} + \vec{b}\} + q\{(2x+1)\vec{a} - \vec{b}\} = 0$$

$$\therefore p(x-2) = -q(2x+1) \text{ and } p\vec{b} = q\vec{b} \quad (\text{since } \vec{a}, \vec{b} \text{ are non-collinear})$$

$$\therefore p(x-2) = -p(2x+1)$$

$$\Rightarrow x = \frac{1}{3}.$$

62. If in  $\triangle ABC$ ,  $\angle A = \sin^{-1}(x)$ ,  $\angle B = \sin^{-1}(y)$  and  $\angle C = \sin^{-1}(z)$ , then  $x\sqrt{1-y^2}\sqrt{1-z^2} + y\sqrt{1-x^2}\sqrt{1-z^2} + z\sqrt{1-x^2}\sqrt{1-y^2}$  is equal to

- (A)  $xyz$   
(B)  $x+y+z$   
(C)  $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$   
(D) None of these

Ans. A

Sol. In a  $\triangle ABC$

$$\sin(A+B+C) = 0$$

$$= \sin A \cos B \cos C + \sin B \cos C \cos A + \sin C \cos A \cos B - \sin A \sin B \sin C$$

$$\Rightarrow xyz = x\sqrt{(1-y^2)(1-z^2)} + y\sqrt{(1-x^2)(1-z^2)} + z\sqrt{(1-x^2)(1-y^2)}$$

63. The value of  $a$  for which the equation  $4\operatorname{cosec}^2(\pi(a+x)) + a^2 - 4a = 0$  has a real solution, is

- (A)  $a = 1$   
 (B)  $a = 2$   
 (C)  $a = 10$   
 (D) None of these

Ans. B

Sol.  $4[1 + \cot^2 \pi(a+x)] + a^2 - 4a = 0$   
 $4\cot^2 \pi(a+x) + (a-2)^2 = 0$   
 $\Rightarrow a-2 = 0$  and  $\cot^2 \pi(a+x) = 0$   
 $\Rightarrow a = 2.$

64. If  $a \sin x + b \cos(C+x) + b \cos(C-x) = \alpha$ , then the minimum value of  $|\cos C|$  is

- (A)  $\sqrt{\frac{\alpha^2 - a^2}{b^2}}$   
 (B)  $\sqrt{\frac{\alpha^2 - a^2}{4b^2}}$   
 (C)  $\sqrt{\frac{\alpha^2 - a^2}{16b^2}}$   
 (D) None of these

Ans. B

Sol. We have  $a \sin x + 2b \cos C \cos x = \alpha$

The maximum value of  $a \sin x + 2b \cos C \cos x$  is  $\sqrt{a^2 + 4b^2 \cos^2 C}$  which increases as  $|\cos C|$  increases. So the minimum value of  $|\cos C|$  will correspond to the situation when  $\sqrt{a^2 + 4b^2 \cos^2 C}$  is just equal to  $\alpha$ , so as to give a real solution of the equation.

$$\therefore \sqrt{a^2 + 4b^2 \cos^2 C} = \alpha$$

$$\cos^2 C = \frac{\alpha^2 - a^2}{4b^2}; |\cos C| = \sqrt{\frac{\alpha^2 - a^2}{4b^2}}.$$

65. If  $5a + 4b + 20c = t$ , then the value of  $t$  for which the line  $ax + by + c - 1 = 0$  always passes through a fixed point is

- (A) 0  
 (B) 20  
 (C) 30  
 (D) None of these

Ans. B

Sol. Equation of line  $\frac{ax}{c-1} + \frac{by}{c-1} + 1 = 0$  has two independent parameters. It can pass through a fixed point if it contains only one independent parameter. Now there must be one relation between  $\frac{a}{c-1}$  and  $\frac{b}{c-1}$  independent of a, b and c so that  $\frac{a}{c-1}$  can be expressed in terms of  $\frac{b}{c-1}$  and straight line contains only one independent parameter. Now that given relation can be expressed as  $\frac{5a}{c-1} + \frac{4b}{c-1} = \frac{t-20c}{c-1}$ .  
Now RHS be independent of c if  $t = 20$ .

66. Consider the equation  $x^2 + x - n = 0$ , where n is an integer lying between 1 to 100. Total number of different values of 'n' so that the equation has integral roots is

- (A) 6
- (B) 4
- (C) 9
- (D) None of these

Ans. C

Sol.  $x^2 + x - n = 0$ , discriminant  $= 1 + 4n = \text{odd number} = D(\text{say})$   
Now given equation would have a integral solution if D is a perfect square.  
Let  $D = (2\lambda + 1)^2 \Rightarrow n = \lambda + \lambda^2 = \lambda(\lambda + 1) = \text{even number}$   
 $\Rightarrow n$  can be 2, 6, 12, 20, 30, 42, 56, 72, 90.

67.  $\sum_{r=1}^n \frac{r^2 - r - 1}{r + 1!}$  is equal to

- (A)  $\frac{n}{(n+1)!}$
- (B)  $\frac{-1}{(n+1)(n-1)!}$
- (C)  $\frac{n}{(n+1)!} - 1$
- (D) None of these

Ans. B

Sol.  $\sum_{r=1}^n \frac{r^2 - r - 1}{r + 1!} = \sum_{r=1}^n \left( \frac{r-1}{r!} - \frac{r}{(r+1)!} \right) = -\frac{n}{(n+1)!}$ .

68. Two events A and B have probabilities 0.25 and 0.50 respectively. The probability that both A and B occur simultaneously is 0.14. Then the probability that neither A nor B occurs is

(A) 0.39  
 (B) 0.25  
 (C) 0.11  
 (D) None of these

Ans. A

Sol.  $P(A) = .25$ ,  $P(B) = .50$ ,  $P(A \cap B) = .14$   
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$   
 $= .25 + .50 - .14 = .61$   
 $P(\overline{A \cap B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - .61 = .39$

69. The number of positive integral solutions of the equation  $x_1 x_2 x_3 = 60$  is

(A) 54  
 (B) 27  
 (C) 81  
 (D) None of these.

Ans. A

Sol. Here  $x_1 x_2 x_3 = 2^2 \times 3 \times 5$   
 Let number of two's given to each of  $x_1, x_2, x_3$  be a, b, c.  
 Then  $a + b + c = 2$ ,  $a, b, c \geq 0$   
 The number of integral solutions of this equations is equal to coefficient of  $x^2$  in  $(1 - x)^{-3}$  i.e.  ${}^4C_2$   
 i.e. the available 2 two's can be distributed among  $x_1, x_2$  and  $x_3$  in  ${}^4C_2 = 6$  ways.  
 Similarly, the available 1 three can be distributed among  $x_1, x_2, x_3$  in  ${}^3C_1 = 3$  ways.  
 (= coefficient of x in  $(1 - x)^{-3}$ )  
 $\therefore$  Total number of ways =  ${}^4C_2 \times {}^3C_1 \times {}^1C_0 = 6 \times 3 \times 1 = 18$  ways.

70. In a bag there are 15 red and 5 white balls. Two balls are chosen at random and one is found to be red. The probability that the second one is also red is

(A)  $\frac{12}{19}$   
 (B)  $\frac{13}{19}$   
 (C)  $\frac{14}{19}$   
 (D)  $\frac{15}{19}$

Ans. C

Sol. Probability that out of remaining balls the one that is red is  $= \frac{{}^{14}C_1}{{}^{19}C_1} = \frac{14}{19}$

71. Let S be the set of 6-digit numbers  $a_1a_2a_3a_4a_5a_6$  (all digits distinct) where  $a_1 > a_2 > a_3 > a_4 < a_5 < a_6$ . Then  $n(S)$  is equal to

- (A) 210
- (B) 2100
- (C) 4200
- (D) 420

Ans. B

Sol. First, 6 distinct digits can be selected in  ${}^{10}C_6$  ways. Now the position of smallest digit in them is fixed i.e. position 4. Of the remaining 5 digits, two digits can be selected in  ${}^5C_2$  ways. These two digits can be placed to the right of 4<sup>th</sup> position in one way only. The remaining three digits to the left of 4<sup>th</sup> position are in the required order automatically.  
So  $n(S) = {}^{10}C_6 \times {}^5C_2 = 210 \times 10 = 2100$ .

72. If the normals at the end points of a variable chord PQ of the parabola  $y^2 - 4y - 2x = 0$  are perpendicular, then the tangents at P and Q will intersect at

- (A)  $x + y = 3$
- (B)  $3x - 7 = 0$
- (C)  $y + 3 = 0$
- (D)  $2x + 5 = 0$

Ans. D

Sol. Since normals at P and Q are perpendicular, the tangents at P and Q will also be perpendicular but any two perpendicular tangents of a parabola always intersect on its directrix. The parabola is  $(y - 2)^2 = 2(x + 2)$ . So its directrix is  $2x + 5 = 0$

73. If A is a skew-symmetric matrix of order 3, then matrix  $A^4$  is

- (A) skew-symmetric
- (B) symmetric
- (C) diagonal
- (D) None of these

Ans. B

Sol. We have  $A' = -A$  and  $(A^4)' = (A \cdot A \cdot A \cdot A)' = A'A'A'A' = (-A)(-A)(-A)(-A) = (-1)^4 A^4 = A^4$   
Hence (B) is the correct answer.

74. Let  $f(x) = \begin{cases} \sin^2 x, & x \text{ rational} \\ -\sin^2 x, & x \text{ irrational} \end{cases}$ . Then set of points, where  $f(x)$  is continuous is

- (A)  $\left\{(2n+1)\frac{\pi}{2}, n \in I\right\}$   
(B) null set  
(C)  $\{n\pi, n \in I\}$   
(D) set of all rational numbers

Ans. C

Sol. If  $a$  is any number for which  $\sin a \neq 0$ , then we don't get any fix tendency of  $f(x)$  to approaches where  $x \rightarrow a$   
so  $\lim_{x \rightarrow a} f(x)$  does not exist  
but if  $\sin a = 0$ ,  $\lim_{x \rightarrow a} f(x) = 0$

75.  $\int \frac{x^{7/2}}{\sqrt{x^9+1}} dx$  equals to

- (A)  $\sin^{-1} \sqrt{x^2+9} + c$   
(B)  $\frac{1}{2} \ln \left( \frac{x^9+1}{x^9-1} \right) + c$   
(C)  $\frac{2}{9} \ln \left( x^{9/2} + \sqrt{x^9+1} \right) + c$   
(D) None of these

Ans. C

Sol. Put  $x^{9/2} = t$  then  $\frac{9}{2} x^{7/2} dx = dt$ , So given integral reduces to

$$\frac{2}{9} \int \frac{dt}{\sqrt{t^2+1}} = \frac{2}{9} \ln \left( t + \sqrt{t^2+1} \right) + c = \frac{2}{9} \ln \left( x^{9/2} + \sqrt{x^9+1} \right) + c$$

76. Four points are such that the line joining any two points is perpendicular to the line joining other two points. If three points out of these lie on a rectangular hyperbola then the fourth point will lie on

(A) the same hyperbola  
 (B) conjugate hyperbola  
 (C) one of the directrix  
 (D) one of the asymptotes

Ans. A

Sol. The points are such that one of the points is the orthocentre of the triangle formed by other three points. When the vertices of a triangle lie on a rectangular hyperbola the orthocentre also lies on the same hyperbola.

77. The value of ' $\alpha$ ' so that the range of the function  $y = \frac{x^2 + \alpha}{x + 1}$  is  $\mathbb{R}$ , is

(A)  $(-\infty, -2]$   
 (B)  $(-\infty, -1]$   
 (C)  $(-\infty, 0)$   
 (D) None of these

Ans. B

Sol. Given that  $y = \frac{x^2 + \alpha}{x + 1}$   
 $\Rightarrow x^2 - yx + \alpha - y = 0$   
 $\Rightarrow y^2 - 4(\alpha - y) \geq 0 \Rightarrow y^2 + 4y + 4 \geq 4\alpha + 4$   
 $4\alpha + 4 \leq 0 \Rightarrow \alpha \leq -1.$

78. The length of the major axis of the ellipse  $(5x - 10)^2 + (5y + 15)^2 = \frac{(3x - 4y + 7)^2}{4}$  is

(A) 10  
 (B)  $\frac{20}{3}$   
 (C)  $\frac{20}{7}$   
 (D) 4

Ans. B

Sol.  $(x-2)^2 + (y+3)^2 = \left(\frac{1}{2} \frac{3x-4y+7}{5}\right)^2$  is an ellipse, whose focus is (2, -3), directrix  $3x - 4y + 7 = 0$  and eccentricity is  $\frac{1}{2}$ .

Length of  $\perp$  from focus to directrix is  $\frac{3 \times 2 - 4 \times (-3) + 7}{5} = 5$

$$\frac{a}{e} - ae = 5 \quad \Rightarrow \quad 2a - \frac{a}{2} = 5 \Rightarrow a = \frac{10}{3}$$

So length of major axis is  $\frac{20}{3}$

79. If  $I(n) = \int_0^{\pi/2} \theta \cdot \sin^n \theta d\theta$ ,  $n \in \mathbb{N}$ ,  $n > 3$ , then the value of  $I(n) - \frac{n-1}{n} I(n-2)$  is

(A)  $\frac{1}{n^2}$

(B)  $\frac{1}{n-1}$

(C)  $\frac{1}{n^2-1}$

(D)  $\frac{1}{n+1}$

Ans. A

Sol.  $I(n) = \int_0^{\pi/2} \theta \cdot \sin^n \theta d\theta$

$$\Rightarrow I(n) = \int_0^{\pi/2} \theta \cdot \sin^{n-2} \theta (1 - \cos^2 \theta) d\theta$$

$$= I(n-2) - \int_0^{\pi/2} (\theta \cdot \cos \theta) \cdot \cos \theta \cdot \sin^{n-2} \theta d\theta$$

$$= I(n-2) - \theta \cdot \cos \theta \cdot \frac{\sin^{n-1} \theta}{n-1} \Big|_0^{\pi/2} + \int_0^{\pi/2} [\theta \cdot (-\sin \theta) + \cos \theta] \frac{\sin^{n-1} \theta}{n-1} d\theta$$

$$= I(n-2) - \frac{1}{(n-1)} \int_0^{\pi/2} \theta \cdot \sin^n \theta d\theta + \frac{1}{n-1} \int_0^{\pi/2} \cos \theta \cdot \sin^{n-1} \theta d\theta$$

$$= I(n-2) - \frac{1}{(n-1)} \cdot I(n) + \frac{1}{(n-1)(n)} \cdot \sin^n \theta \Big|_0^{\pi/2}$$

$$\Rightarrow \frac{n}{n-1} I(n) = I(n-2) + \frac{1}{(n-1)(n)}, \Rightarrow I(n) - I(n-2) \cdot \frac{n-1}{n} = \frac{1}{n^2}.$$

80. Let  $x_1y_1z_1$ ,  $x_2y_2z_2$  and  $x_3y_3z_3$  be three 3-digit even numbers and  $\Delta = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$ . Then  $\Delta$  is

- (A) divisible by 2 but not necessarily by 4
- (B) divisible by 4 but not necessarily by 8
- (C) divisible by 8
- (D) none of these

Ans. A

Sol.  $\Delta = \begin{vmatrix} 100x_1 + 10y_1 + z_1 & y_1 & z_1 \\ 100x_2 + 10y_2 + z_2 & y_2 & z_2 \\ 100x_3 + 10y_3 + z_3 & y_3 & z_3 \end{vmatrix} = \begin{vmatrix} 2A & y_1 & z_1 \\ 2B & y_2 & z_2 \\ 2C & y_3 & z_3 \end{vmatrix}$ , where  $A, B, C \in \mathbb{I}$   
 $= 2 \begin{vmatrix} A & y_1 & z_1 \\ B & y_2 & z_2 \\ C & y_3 & z_3 \end{vmatrix}$ , which is divisible by 2 but not necessarily by 4 or 8.

### SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If the line  $y = \sqrt{3}x$  cuts the curve  $x^4 + ax^2y + bxy + cx + dy + 6 = 0$  at A, B, C and D, then OA.OB.OC.OD (where O is the origin) is

Ans. 00096.00

Sol. The line  $y = \sqrt{3}x$  can be written as  $x = \frac{r}{2}$ ,  $y = \frac{r\sqrt{3}}{2}$ . If this line cuts the given curve, then

$$\frac{r^4}{16} + \frac{ar^3\sqrt{3}}{8} + \frac{br^2\sqrt{3}}{4} + \frac{cr}{2} + \frac{dr\sqrt{3}}{2} + 6 = 0.$$

$$\text{Therefore OA. OB.OC.OD} = |r_1| |r_2| |r_3| |r_4| = |r_1 r_2 r_3 r_4| = 96$$

82. The value of 'a' for which the equation  $(a^2 + 4a + 3)x^2 + (a^2 - a - 2)x + (a + 1)a = 0$  has more than two roots is  $-k$ , then the value of  $k+2020$  is

Ans. 02021.00

Sol.  $(a^2 + 4a + 3)x^2 + (a^2 - a - 2)x + (a + 1)a = 0 \Rightarrow (a + 1)(a + 3)x^2 + (a + 1)(a - 2)x + (a + 1)a = 0$   
 $(a + 1)[(a + 3)x^2 + (a - 2)x + a] = 0$   
 if  $a = -1$ , then quadratic have more than two roots.

83. Equation of the normal to the curve  $x^2 = 4y$  which passes through the point (1, 2) is  $x + y = k-12$ , then  $k =$

Ans. 00015.00

Sol. Equation of the normal to the curve  $x^2 = 4y$  is  $x = my - 2m - m^3$ , passes through the point (1, 2)  $\Rightarrow m = -1$ .

84. If the value of the sum  $29\binom{30}{0} + 28\binom{30}{1} + 27\binom{30}{2} + \dots + 1\binom{30}{28} + 0\binom{30}{29} - \binom{30}{30}$  where  $\binom{n}{r} = {}^nC_r$ , is equal to  $k2^{32}$ , then the value of  $k$  is equal to

Ans. 00003.50

Sol.  $S = 29\binom{30}{0} + 28\binom{30}{1} + 27\binom{30}{2} + \dots + 1\binom{30}{28} + 0\binom{30}{29} - \binom{30}{30}$

$$\therefore 2S = 28({}^{30}C_0 + {}^{30}C_1 + {}^{30}C_2 + \dots + {}^{30}C_{30}) = 28(2^{30})$$

$$\text{Hence, } S = 14 \times 2^{30} = \frac{7}{2} \times 2^{32} \Rightarrow k = \frac{7}{2}$$

85. Let  $A = [a_{ij}]$  be a  $3 \times 3$  invertible matrix. If determinant value of matrix  $A$  is 3, then the value of  $\det((\text{adj} A^T)^T) + \det((\text{adj} A^{-1})^{-1})$  (where  $\det(B)$  denotes determinant value of matrix  $B$ )

Ans. 00018.00

Sol.  $\det((\text{adj} A^T)^T) = (\det A)^2 = 9$   
 $\det((\text{adj} A^{-1})^{-1}) = (\det A)^2 = 9$

86. If  $z_1, z_2, z_3$  are complex number such that  $|z_1| = 2, |z_2| = 3, |z_3| = 4$  then maximum value of  $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$  is

Ans. 00087.00

Sol.  $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$   
 $= 2(|z_1|^2 + |z_2|^2 + |z_3|^2) - (z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1)$   
 $= 58 - (z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1) \dots\dots(1)$

$$\text{Now } |z_1 + z_2 + z_3|^2 \geq 0$$

$$\Rightarrow |z_1|^2 + |z_2|^2 + |z_3|^2 + z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1 \geq 0$$

$$\Rightarrow -(z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1) \leq 29 \dots\dots(2)$$

from (1) and (2)

$$\text{maximum value} = 58 + 29 = 87.$$

87. A point  $P(x, y)$  is said to be integral point iff both  $x, y$  are integers. Then number of integer points inside or on the circle  $x^2 + y^2 = 9$  is

Ans. 00029.00

Sol. Let  $(n, m)$  be a integer point, then  
 $n^2 + m^2 \leq 9$

if  $n^2 = 9, m^2 = 0 \Rightarrow 2 \times 1 = 2$  points  
 if  $n^2 = 4, m^2 = 0, 1, 4 \Rightarrow 2 \times 5 = 10$  points  
 if  $n^2 = 1, m^2 = 0, 1, 4 \Rightarrow 2 \times 5 = 10$  points  
 if  $n^2 = 0, m^2 = 0, 1, 4, 9 \Rightarrow 1 \times 7 = 7$  points  
 so in total, we have 29 points, which are integral points.

88. Let  $(1+x)^n = 3 + \frac{8}{3} + \frac{80}{3^3} + \frac{240}{3^4} + \dots$ ,  $x, n \in \mathbb{R}$ . Then the sum of the series  $3 + \frac{8}{3} + \frac{80}{3^3} + \frac{240}{3^4} + \dots$  is

Ans. 00027.00

Sol.  $(1+x)^n = 3 + \frac{8}{3} + \frac{80}{3^3} + \frac{240}{3^4} + \dots = 1 + nx + \frac{n(n-1)x^2}{2} + \dots$   
 On comparison,  $n = -3$  and  $x = \frac{-2}{3}$ .

89. If  $a_i, i = 1, 2, 3, 4$  be four real numbers of the same sign, then the minimum value of  $\sum \frac{a_i}{a_j}$ ,  $i, j \in \{1, 2, 3, 4\}, i \neq j$ , is

Ans. 00012.00

Sol.  $\sum \frac{a_i}{a_j}, i, j \in \{1, 2, 3, 4\}, i \neq j$  has a total of  $4 \times 4 - 4 = 12$  terms i.e. 6 pairs, each pair of the type  $\left( \frac{a_i}{a_j} + \frac{a_j}{a_i} \right)$ .  
 Now  $\left( \frac{a_i}{a_j} + \frac{a_j}{a_i} \right) \geq 2$  (as  $\frac{a_i}{a_j} > 0$ )  
 $\Rightarrow \sum \frac{a_i}{a_j} \geq 2 \times 6 \Rightarrow \sum \frac{a_i}{a_j} \geq 12$ .

90. Let  $N$  be any four digit number say  $x_1 x_2 x_3 x_4$ . Then maximum value of  $\frac{N}{x_1 + x_2 + x_3 + x_4}$  is equal to

Ans. 01000.00

Sol.  $\frac{N}{x_1 + x_2 + x_3 + x_4} = \frac{1000x_1 + 100x_2 + 10x_3 + x_4}{x_1 + x_2 + x_3 + x_4}$   
 $= 1000 - \frac{(900x_2 + 990x_3 + 999x_4)}{(x_1 + x_2 + x_3 + x_4)} \Rightarrow \text{maximum value of } \frac{N}{x_1 + x_2 + x_3 + x_4} = 1000$